

Detection of Reticuloendotheliosis Virus in Muscovy Ducks, Wild Turkeys, and Chickens in Brazil

Giovana S. Caleiro,¹ Cristina F. Nunes,¹ Paulo R. Urbano,¹ Karin Kirchgatter,² Jansen de Araujo,³ Edison Luiz Durigon,³ Luciano M. Thomazelli,³ Brittany M. Stewart,⁴ Dustin C. Edwards,⁴ and Camila M. Romano^{1,5} ¹Instituto de Medicina Tropical de São Paulo e Hospital das Clínicas da Faculdade de Medicina, HCFMUSP (LIM52) Av. Dr. Enéas Carvalho de Aguiar, 470, CEP 05403-000, São Paulo/SP, Brasil, Universidade de São Paulo, São Paulo, Brazil; ²Instituto de Medicina Tropical de São Paulo, SUCEN, Av. Dr. Enéas Carvalho de Aguiar, 470, CEP 05403-000, São Paulo/SP, Brazil; ³Laboratório de Virologia Clínica e Molecular, Instituto de Ciências Biomédicas (ICB), Universidade de São Paulo, Av. Prof. Lineu Prestes, nº 2415CEP 05508-900, Butantã, São Paulo/SP, Brazil; ⁴Department of Biological Sciences, Tarleton State University, Box T-0100, Stephenville, Texas 76402; ⁵Corresponding author (email: cmromano@usp.br)

ABSTRACT: Reticuloendotheliosis retroviruses (REVs) are known to cause immunosuppressive and oncogenic disease that affects numerous avian species. Reticuloendotheliosis retroviruses are present worldwide and recently have been reported in South America with cases of infected commercial flocks in Argentina. We surveyed for the presence of REV in birds from a state in the northern region of Brazil using real-time PCR. We report here the presence of REV in Brazil, detected in Muscovy Ducks (*Cairina moschata*), Wild Turkeys (*Meleagris gallopavo*), and chickens (*Gallus gallus*) at a relatively high prevalence (16.8%). Phylogenetic analysis indicated a close relationship of these strains to variants in the United States. This study provides evidence of REV in the Amazon biome and provides a baseline for future surveillance of the virus in the region and throughout Brazil.

Key words: Avian retrovirus, Brazil, reticuloendotheliosis virus, REV.

Reticuloendotheliosis viruses (REVs) are a group of immunosuppressive and sometimes oncogenic retroviruses that affect numerous species of birds, including waterbirds (Anseriformes), game birds (Galliformes), and perching birds (Passeriformes; Nair et al. 2013). Reticuloendotheliosis viruses has previously affected commercial poultry and has been a recurring obstacle in the conservation of endangered species in some countries, such as Attwater's Prairie Chicken (*Tympanuchus cupido attwateri*; Ferro et al. 2017). In Brazil, the impact of REV in endangered species is not known. Among the birds that are putatively susceptible to REV infection are the critically endangered anseriform Brazilian Merganser (*Mergus octocetaceus*) and the

galliform Red-billed Curassow (*Crax blumembachii*; Ministry of the Environment, 2014). Within vaccinated flocks, REV infection can lead to decreased efficacy of vaccinations for avian influenza virus, Newcastle disease virus, Marek's disease virus (MDV), and turkey herpesvirus due to the reduced humoral response resulting from immunosuppression (Sun et al. 2009). The virus was first identified in the US in 1958, and later in China, Taiwan, Australia, Argentina, and Canada (Singh et al. 2003; MacDonald et al. 2018). The prevalence of REV varies from 0% to 50% depending on the region, setting, and avian species (Jiang et al. 2013; Ferro et al. 2017; Stewart et al. 2019).

Evidence of REV in South America was recently demonstrated in fowlpox-vaccinated commercial poultry flocks in Argentina (Buscaglia 2013). Reticuloendotheliosis virus proviral DNA has been detected in the genomes of the attenuated MDV vaccine (MD-2 strain) and the field and vaccine strains of fowlpox viruses (FWPV-REV; Isfort et al. 1992; Hertig et al. 1997). Vaccine-integrated REV can be either infectious or noninfectious (Hertig et al. 1997; Moore et al. 2000). The Brazilian coast serves as an important stopover site for migratory birds coming from the northern hemisphere. Several species migrate during the austral winter from Argentina and Chile to central Bolivia and Brazil (Somenzari et al. 2018). In addition, several birds, including Passeriformes, migrate from the US, Mexico, and Central America to northern Brazil (Somenzari et al. 2018). However, no studies have surveyed for the presence of REV in

Brazil. With more than 2,000 species of wild birds present in the country, many of which are endangered, determining the presence of REV and establishing a baseline prevalence rate could be of value for future conservation efforts in the Amazon biome.

During 2005–2006, a total of 441 samples were collected (blood and pooled cloacal or orotracheal swabs) near eight different cities in the northeastern region of Pará state, Brazil. Most of the samples were from Muscovy Ducks (*Cairina moschata*, $n=379$), and the remaining samples were from Wild Turkeys (*Meleagris gallopavo*, $n=41$) and chickens (*Gallus gallus*, $n=21$). Orotracheal or cloacal swab samples were obtained from a previous avian influenza virus study (Thomazelli et al. 2012). All samples were collected from live animals with no apparent illness using simple traps (drop nets) or manual restraint. The birds were free-living or semi-domesticated. The semidomesticated birds were kept outdoors, so contact with other birds was frequent. For the present study, cDNA previously used in Thomazelli (2014) was investigated for the presence of REV genetic material. As an internal control, a conventional PCR method for avian mitochondrial DNA was performed using cytochrome *b* primers (Kocher et al. 1989). The presence of REV proviral cDNA was detected using primers specific to the *gp90* gene (*env*) and real-time PCR (Li et al. 2013). Reactions were prepared using 2.5 mM of each primer, 7.5 μ L of water, 12.5 μ L of MasterMix SYBR® Green (LifeTechnologies®, São Paulo, SP, Brazil), and 3 μ L of purified cDNA. Plasmids containing genes of interest were constructed from commercially synthesized inserts (GenScript, Piscataway, NJ, USA) to serve as positive controls. In addition, partial fragments of LTR-U5, Gag, and envelope from REV-positive samples were amplified and sequenced by the Sanger method using previously described primers (Singh et al. 2003; Barbosa et al. 2007; Li et al. 2013). For the envelope region, an additional primer was designed (gp90-7242R 5'-GCCAGTATGCA CAGCCCTATCCA-3').

Of the 441 samples tested and amplified, REV PCR products were detected in 74 samples (17%). Infected individuals included 65 Muscovy Ducks, six Wild Turkeys, and two chickens. The REV-positive samples came from birds in Marabitaná, Vila Maracajá, and Marajó. Unfortunately, only viruses obtained from Muscovy Ducks were successfully amplified and sequenced. Sequencing reads were inspected for quality and consensus sequences were built using the CLC Genomic Workbench v5 (Qiagen, Redwood City, California, USA) and submitted to GenBank (accession numbers MG953804–MG953809 and MN389236–MN389242). Consensus sequences of partial LTR/gag (808 base pairs), gag (1,168 base pairs), and envelope (~600 base pairs) were aligned and compared to reference strains and other sequences, retrieved from GenBank, that represented worldwide REV ($n=16$). Phylogenetic trees were reconstructed under the maximum-likelihood method in PhyML 3.0 software (Guindon et al. 2010) using K80 as the best nucleotide substitution model for all fragments as determined by jModeltest (Darriba et al. 2012). Only sequences built with high quality reads (five sequences from envelope region and seven from LTR/gag—Phred >30) were used for phylogenetic reconstructions. Together, sequences from nine different isolates were used to build either envelope or LTR/gag trees or both. Genetic distance (*p*-distance) between the main REV strains and one representative Brazilian REV sample was estimated for LTR/gag, gag, and envelope partial fragments. To establish the phylogenetic relationships among Brazilian and worldwide REV, phylogenetic trees were reconstructed using LTR/gag and envelope fragments representing Brazilian and worldwide REV genomic sequences ($n=16$; Fig. 1). Brazilian REV clustered together with sequences sampled in the US, such as Attwater's Prairie Chicken reference strain APC-566 (Barbosa et al. 2007). Genetic distance at the nucleotide level (*p*-distance) between the main REV strains and one representative Brazilian REV (REV_2036_BR) also agreed that Brazilian REV is more closely related to the US samples representative of subtype 3, such as MD-2 and APC-566, rather than to

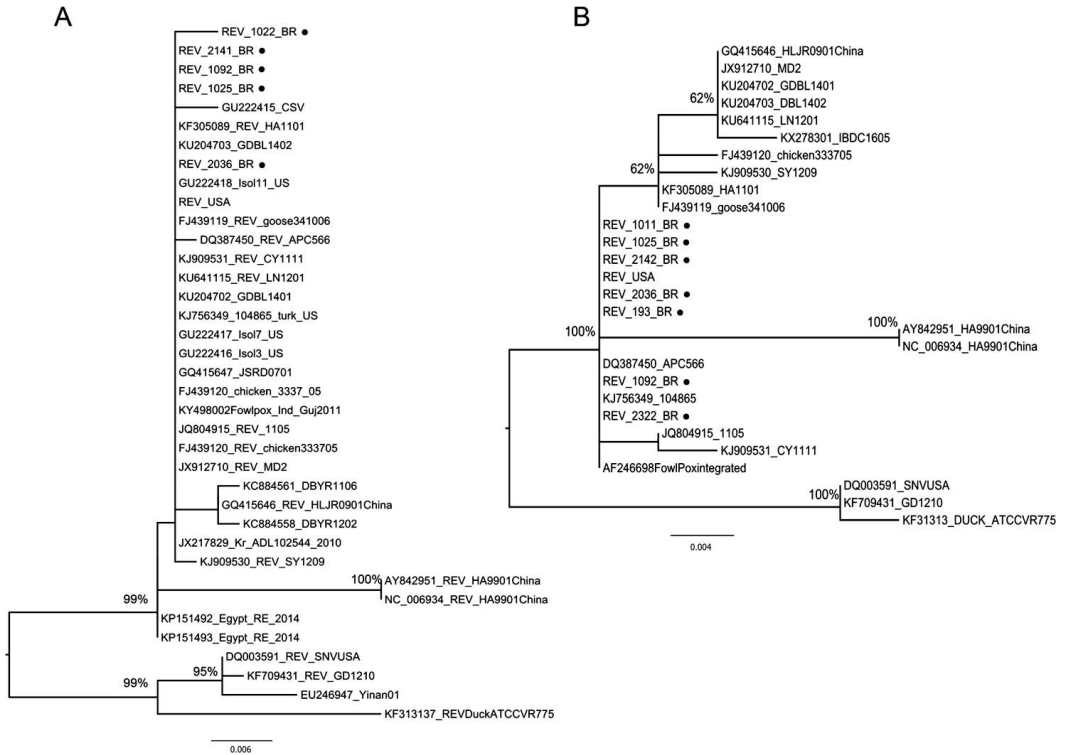


FIGURE 1. Maximum-likelihood phylogenetic reconstruction showing the relationships between reticuloendotheliosis retrovirus from Brazil and other isolates. Brazilian sequences (black closed circles) are from Muscovy Ducks (*Cairina moschata*). The trees were made with partial sequences from reticuloendotheliosis retrovirus *env* A) and *LTR/gag* B) available at GenBank.

spleen necrosis virus (SNV) strain or China isolates (Fig. 1 and Table 1).

Pará state is located in the northern coastal region of Brazil and has a tropical rainforest climate. Because of its geographic location and ecosystem, Pará is a stopover site for shorebirds coming from the northern hemi-

sphere during migration. In addition, several species of migratory birds that have Argentina and the US as their final destinations pass through Brazil during their flight. The similarity between Brazilian REV and the US viruses suggests that the US could be the source of REV detected in Brazil. However,

TABLE 1. Genetic identity (p-distance) in percentage at nucleotide level between Brazilian reticuloendotheliosis retrovirus (REV) strain (REV_2036_BR) and reference strains from GenBank.

REV reference	GenBank ID	REV_2036_BR		
		LTR/gag (%)	Gag (%)	Env (%)
APC-566	DQ387450	99.8	100	99.8
SNV	DQ003591	96	97.5	96.4
MD-2	JX912710	99.4	100	100
FPV	AF246698	99.6	100	100
CSV	DQ237905	NA ^a	NA	99.5
HA9901 (China)	AY842951	96	98	97.6

^a NA = sequence not available.

no sequence data are available for viruses originating in Argentina. As such, the source of Brazilian REV cannot be definitively determined. Representative strains of REV include the defective REV-T, the nondefective REV-A, SNV, duck infectious anemia, and chicken syncytial virus (CSV), and the MD-2 and FWPV vaccine-integrated strains. As there is relatively little genetic variation among REV strains, it is difficult to determine the origin and classification of novel isolates (Bohls et al. 2006). In our analysis, Brazilian REV differs only 0.0–0.5% from both FWPV and MD-2 integrated REVs and nonintegrated APC-566 and CSV. We inspected for point mutations, which have been characterized as specific for FWPV-REV (Tadese et al. 2008). According to the genetic pattern observed, the Brazilian REV samples are all non-FWPV-integrated strains. However, not all FWPV-REV envelope sequences available on GenBank (KY498002, AF246698, and JX217830) contain this signature, indicating that some genetic variability among integrated strains exists and that determining the integration status based on a few point mutations is inaccurate (Tadese et al. 2008). Therefore, it remains unknown if Brazilian REV is present as infectious particles or is instead integrated within a large DNA virus genome.

Reticuloendotheliosis viruses can infect a number of species, including captive and wild perching birds, game birds, and waterbirds. Although we detected REV in Muscovy Ducks, Wild Turkeys, and chickens, we could amplify and sequence the virus only from ducks. It is possible that REV viral loads were lower in other species, precluding our ability to generate large amplicons. Muscovy Ducks are native to Mexico and Central and South America, but populations of Muscovy Ducks also reside in the US, mainly in Florida and southern Texas. These birds are essentially nonmigratory or irregular migrants without established migration patterns, only migrating short distances to avoid dry weather and fluctuating water conditions. Chickens and turkeys, which are frequently associated with REV infections, are also nonmigratory birds. Therefore, we consider it unlikely that the

positive birds found in Brazil were infected elsewhere.

Here, we describe the presence of REV in Brazil and present sequences of REV provirus from South America. The REV found in Brazil is similar to other common circulating strains, including those found in the US. Additionally, considering that REV was present in samples from three of the five regions collected from Pará in 2005–2006 (located 200–500 km apart from each other), it is very likely that the virus has spread to other states in Brazil. We hypothesize that the virus could be carried by migratory birds that stop over in the northern part of the country on their way to and from North America. These results suggest the need for additional studies to further determine the prevalence and genetic variability of REV in Brazilian wild and captive birds and also to perform risk assessment studies dedicated to free-range and captive commercial avian species. In addition to REV, it will be also important to survey for the presence of lymphoproliferative disease virus in the future, as it is an oncogenic retrovirus that infects domestic and wild turkeys and chickens across the US (Alger et al. 2017).

All procedures involving wild and captive birds were approved by the Animal Ethics Committee from the Instituto de Medicina Tropical de São Paulo under protocol 000283A, and licensed by the Ministério do Meio Ambiente–MMA at the Instituto Chico Mendes de Conservação da Biodiversidade (ICM-Bio/SISBIO) under protocols 34605-7 45527-2, 54616-1, and 201/2006 CGFAU. This work was supported by Fundação de Amparo a Pesquisa do Estado de São Paulo (FAPESP Project #2015/05958-3) and by the Programa de fomento às atividades de lideranças científicas dos LIMs do Hospital das Clínicas da FMUSP (PROFAP-LIM/HCFMUSP) number 20/2019. G. Caleiro holds a CAPES scholarship.

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Submitted for publication 2 April 2019.

Accepted 4 September 2019.